

Day 2 Mastery Quiz: Variables, Expressions, and State — Instructor Key

Current value → planned value → approval checkpoint → state-history explanation

Instructor key. Same layout as student copy; solutions filled in response boxes. Print duplex on one sheet.

Name		Section		Date	
Score	____/10		Mastery check		secure / developing / needs support

A pump-calibration team is checking a flow record before it enters a lab log. The notebook stores a current recorded flow, calculates planned corrections, and records a planned value only after approval. Trace the current state after each cell: planned is not current until an assignment makes it current.

1 Reference code for Items 1–10

[predict](#) [trace](#)

```
1 # A: baseline record
2 pump_id = "P-18"
3 target_flow_lpm = 20.0
4 flow_lpm = 18.5
5 gap_lpm = target_flow_lpm - flow_lpm
6 # B: first planned correction
7 adjust_lpm = 1.0
8 planned_flow_lpm = flow_lpm + adjust_lpm
9 planned_gap_lpm = target_flow_lpm - planned_flow_lpm
10 # C: approve first plan
11 flow_lpm = planned_flow_lpm
12 gap_lpm = planned_gap_lpm
13 # D: fine adjustment, not recorded yet
14 adjust_lpm = 0.5
15 planned_flow_lpm = flow_lpm + adjust_lpm
16 gap_lpm = target_flow_lpm - planned_flow_lpm
17 # E: record fine adjustment after approval
18 flow_lpm = planned_flow_lpm
19 gap_lpm = target_flow_lpm - flow_lpm
```

2 Items 1–2: identify current and planned roles

[recognize](#)

Pt.	Prompt	My response
1	Which variable stores the current recorded pump flow?	Key: flow_lpm
1	Which variable stores a planned flow value before approval?	Key: planned_flow_lpm

3 Items 3–6: trace state after key checkpoints

[trace](#) [explain](#)

Pt.	Prompt	My response
1	After Cell A, what are the current values of flow_lpm and gap_lpm?	Key: flow_lpm = 18.5; gap_lpm = 1.5.
1	After Cell B, what are the current recorded flow and the planned flow?	Key: Current recorded flow remains 18.5; planned flow is 19.5.
1	Which assignment in Cell C makes the first planned flow become the current recorded flow?	Key: flow_lpm = planned_flow_lpm
1	After Cell D, what are flow_lpm, planned_flow_lpm, and gap_lpm?	Key: flow_lpm = 19.5; planned_flow_lpm = 20.0; gap_lpm = 0.0.

4 Items 7–8: debug a lab-log claim

[debug](#) [test](#)

Claim to check after Cell D

The pump flow is 20.0 L/min because `gap_lpm` is now 0.0.

Pt.	Prompt	My response
1	What is unsafe or incomplete in the claim? Use current/planned state and checkpoint language.	Key: After Cell D, 20.0 is planned, not current. Current <code>flow_lpm</code> is still 19.5; the zero gap was calculated from planned flow.
1	Write one Python line you could run after Cell D to check the current recorded flow without changing any variable.	Key: <code>print(flow_lpm)</code> or bare <code>flow_lpm</code> .

5 Item 9: complete the approval assignment

[implement](#) [trace](#)

After Cell D, the team approves the planned fine adjustment. Complete the assignment line that records the planned value as the current value.

```
flow_lpm = -----
```

Pt.	Prompt	My response
1	Write the expression or variable name that belongs on the right side.	Key: <code>planned_flow_lpm</code> ; full line: <code>flow_lpm = planned_flow_lpm</code> .

6 Item 10: transfer the state-history rule

[transfer](#) [explain](#)

Pt.	Prompt	My response
1	In one sentence, explain why a lab log should name the checkpoint before copying a flow value from this notebook.	Key: The checkpoint shows whether a value was only planned or actually assigned as current recorded flow.

Support route, not scored

My issue is mostly: setup / Python syntax / tracing / arithmetic / state history / assignment order / confidence / time.

My next small action is: rerun one cell / rebuild the trace table / ask a partner / ask instructor / try a recovery version.